

AI-DRIVEN PLAYWRIGHT SYNTHETIC MONITORING PLATFORM

Cloud-Native • Kubernetes-Native • AI-Enabled Reference Architecture

SOURCE CONTROL & CI/CD PIPELINE

GitHub Repository — master branch

Developers push commits / merge PRs
Branch-protection + status checks enabled

CI Build Pipeline (GitHub Actions)

Build → Unit Tests → Container Image
Push image to Container Registry

GitOps Deploy (Argo CD)

Syncs manifests to Kubernetes cluster
Progressive delivery / rollback support

Commit Webhook Trigger

push event (ref = master) → HTTPS POST
Signed payload (HMAC secret)

KUBERNETES CLUSTER (EKS / GKE / AKS — multi-zone, autoscaled)

Header color denotes layer type

Ingress Controller + Istio Service Mesh

HPA / KEDA Event-Driven Autoscaling

ArgoCD GitOps Sync

Vault Secrets Manager

Prometheus • Grafana • Loki

OpenTelemetry Distributed Tracing

API Gateway / Webhook Receiver (K8s Ingress)

Validates HMAC signature • filters ref=refs/heads/master
Rate-limited, TLS-terminated endpoint

Event Bus (Kafka / RabbitMQ)

publish Topic: commit.pushed → fan-out to consumers
Durable, replayable event log

Deployment Intelligence AI Agent

microservice • event-driven

Commit Diff Analyzer

Pulls latest vs previous commit
AST / semantic code diff
Flags changed UI routes & components

Build Comparator

Compares latest vs previous build
Bundle size, dependency & config delta

LLM Impact & Risk Engine

Reasons over diff + build delta
Scores deployment risk
Identifies affected user journeys

AI Test Case Generator

Auto-authors Playwright specs (.ts)
Covers new/changed flows
Commits specs to test repo

LLM inference

generated Playwright specs (.spec.ts)

Playwright Execution Engine

kubernetes jobs • horizontally scaled

Test Orchestrator & Scheduler

Triggered by newly generated specs
Schedules K8s jobs per test suite

Test Queue (Redis / Kafka)

Backpressure-aware job queue
Priority scheduling & retries

Distributed Runner Pods

Chromium / Firefox / WebKit
Parallel sharding • HPA/KEDA autoscaled

Artifact Capture Sidecar

Screenshots on each step
Video recording + trace + HAR
Console & network logs

artifacts + results

DATA & STORAGE LAYER

Object Storage (S3 / MinIO)

Screenshots, videos, traces, HAR

Metadata Store (PostgreSQL)

Test runs, results, commit/build history

Cache & Queue (Redis)

Job state, pub/sub, rate limiting

Vector Knowledge Base

Embeddings: code, failures, fixes

insights + embeddings (feedback loop)

on test failure

AI Root Cause Agent

triggered on test.failed

Failure Event Listener

Subscribes to test.failed events
De-dupes flaky-vs-real failures

Log & Artifact Correlator

Aggregates screenshots, video, DOM
Console + network + trace timeline

Multimodal LLM Analysis Engine

Vision + log reasoning
Correlates with commit diff context

Insight Generator

Failure Summary
Root Cause
Suggested Fix + confidence score

LLM inference

results, insights, metadata

BACKEND API & ORCHESTRATION LAYER

API Gateway

REST / GraphQL / WebSocket

AuthN / AuthZ

OIDC SSO • RBAC

Notification Service

Slack • Teams • Email • PagerDuty

Workflow Orchestrator

Argo Workflows / Temporal

REST / GraphQL / WebSocket

Operations Dashboard — Modern AI-Native Frontend (React / Next.js)

served via CDN + API Gateway

Real-Time Monitoring

Live pass/fail, uptime & latency SLAs
Global synthetic-check map

Deployment Intelligence View

Commit diff, risk score
Generated tests per deploy

Root Cause Analysis View

Failure summary + suggested fix
Screenshot / video replay

AI Copilot Chat

Natural-language Q&A
Over test history & incidents

monitors & investigates

Operations / SRE Team